



Hybrid Solar-Electric Cart Innovation for Sustainable Urban Micro-Mobility: Design, Prototyping, and Performance Evaluation of a Solar-Powered MSME Mobility Solution

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Abstract: Low-speed urban micro-mobility can support sustainable last-mile transportation in dense urban environments, particularly where compact vehicle dimensions, cargo capability, and flexible energy access are required. This study presents the design and experimental evaluation of a Hybrid Solar-Electric Cart (HSEC) as a photovoltaic (PV)-assisted urban micro-mobility platform, with micro-, small-, and medium-sized enterprise (MSME) logistics considered as a principal application scenario. A Design-Build-Test-Evaluate (DBTE) engineering framework was applied to prototype development, system integration, and field validation. The HSEC integrates a reinforced steel chassis, a 2 kW 72 V brushless direct current (BLDC) mid-drive motor, dual 72 V 22 Ah LiFePO₄ batteries, a 300 W PV array, battery management system (BMS)-based battery-state monitoring, maximum power point tracking (MPPT)-regulated charging, and automatic battery switching. These functions form a coordinated energy-management architecture that supports source monitoring and propulsion continuity during low-speed operation. Under the reported field conditions, the prototype achieved an average operating range of approximately 40 km per battery, remained functional under combined rider-and-cargo loads of up to 180 kg, completed low-speed turning maneuvers with an approximate radius of 5.6 m, and charged one battery from 30% to 90% state of charge (SOC) in approximately 3.6 h under tropical solar conditions. The findings demonstrate the engineering feasibility of integrating PV charging, dual-battery energy management, and compact cargo mobility within a low-speed transportation platform. The proposed architecture may support last-mile logistics, neighborhood transport, designated vendor routes, campuses, tourism areas, and other low-speed mixed-use environments, subject to further durability, safety, ergonomic, environmental, and techno-economic validation.

Keywords: Urban micro-mobility; Hybrid solar-electric cart; Renewable energy systems; Photovoltaic charging; Sustainable transportation; LiFePO₄ battery; MSME mobility; Prototype engineering

1 Introduction

Urban micro-mobility has emerged as an important component of sustainable transportation systems because it supports low-carbon mobility, improves accessibility, and enhances the operational efficiency of urban economic activities. In developing countries, micro-, small-, and medium-sized enterprises (MSMEs), particularly street vendors, depend heavily on affordable and flexible mobility systems to distribute goods within dense urban environments. Consequently, improving MSME mobility is not only an engineering challenge but also an urban sustainability issue involving transportation planning, public-space management, and renewable-energy utilization [1–3]. Recent research has further demonstrated the potential of photovoltaic (PV)-based charging systems to support electric mobility, particularly in regions with abundant solar resources. In the Indonesian context, PV-powered Electric Vehicle (EV) charging has been shown to offer a technically and economically promising pathway for integrating renewable energy into transportation infrastructure [4]. For street vendors, mobility technology is not only a vehicle-design issue; it is also shaped by the built environment, including narrow access routes, uneven pavements, limited curb space, congestion, exposure to weather, and restrictions on the use of public space. These

spatial conditions create a need for compact, maneuverable, low-emission vehicles that can operate without requiring dedicated charging infrastructure.

In Indonesia, street-vendor activities are commonly managed through local-government arrangements concerning vendor registration, designated trading locations, public-space order, pedestrian circulation, and MSME empowerment. MSME street vendors rely heavily on manual carts or low-efficiency electric modifications that lack adequate energy storage [5], ergonomic design [6], or charging accessibility [6, 7]. Charging infrastructure accessibility is also an important consideration in the broader transition toward electric mobility. Recent research has emphasized that the spatial availability and strategic placement of charging facilities are important for supporting EV accessibility, utilization, and sustainable urban mobility [8]. Consequently, a mobility solution for vendors must be compatible with local spatial governance rather than treated solely as an engineering product. The Hybrid Solar-Electric Cart (HSEC) was therefore conceived for short urban trips, dense market areas, neighborhood corridors, and locations where grid-charging access is limited. Its potential deployment should be coordinated with designated vendor zones, market management, non-motorized transport planning, and local MSME support programs. Therefore, mobility technologies developed for MSMEs should be evaluated not only from engineering performance but also from their compatibility with urban governance, pedestrian safety, and designated vendor-management policies.

Although previous studies have investigated electric micro-vehicles, vehicle-integrated PV systems, battery technologies, and sustainable transportation, relatively few studies have examined their combined application in urban vending carts specifically designed for MSME operations. Existing studies generally focus on individual engineering subsystems, such as PV integration, battery management, or electric propulsion, whereas comprehensive prototype development integrating structural design, renewable-energy harvesting, operational continuity, ergonomics, and urban usability remains limited [9].

The novelty of this study does not lie in introducing an entirely new PV or electric propulsion technology. Instead, it lies in engineering the integration of structural design, PV charging, dual-battery energy management, and operational validation into a single prototype specifically developed for urban MSME mobility.

State-of-the-art analyses further show that previous vehicle-integrated PV systems emphasize rooftop installations, automotive-scale optimization, or macro-transport infrastructures. However, none address the unique constraints of MSMEs, including load variability, mobility through narrow pathways, cost limitations, or high daily operational cycles. State-of-the-art vehicle-integrated PV research has primarily emphasized passenger vehicles, electric cars, lightweight recreational vehicles, and transportation infrastructure. Comparatively little attention has been devoted to renewable-energy-powered cargo platforms supporting informal urban economies, despite their considerable contribution to sustainable urban logistics.

This study therefore aims to: (1) design a structurally reinforced, ergonomic, and energy-autonomous HSEC platform; (2) integrate a dual-battery hybrid solar-electric powertrain with coordinated energy-management functions; (3) empirically evaluate its endurance, load capacity, maneuverability, PV-charging performance, and battery-switching operation; and (4) assess its engineering feasibility as a low-speed urban micro-mobility platform for short-distance and last-mile transportation. Within this transportation context, MSME vending represents an important application scenario rather than the sole intended use of the platform. The HSEC architecture may also be relevant to neighborhood logistics, designated vendor routes, campuses, tourism areas, market-service corridors, and other low-speed mixed-use environments where compact vehicle dimensions, maneuverability, operational continuity, and reduced dependence on fixed charging infrastructure are important. Accordingly, this study positions the HSEC within the broader development of sustainable and intelligent urban micro-mobility systems rather than solely as a renewable-energy cart for MSME operations.

2 Methodology

2.1 Research Design

This study employed a Design–Build–Test–Evaluate (DBTE) engineering framework to guide the systematic development and validation of a HSEC designed for sustainable urban mobility applications. The DBTE framework was selected because it provides a structured engineering methodology that integrates conceptual design, prototype fabrication, experimental verification, and performance evaluation within a continuous development cycle. Rather than focusing solely on component-level performance, the framework enables evaluation of the complete engineering system under practical operating conditions [10].

The Design stage began with field observations and problem identification to determine the operational limitations of conventional street-vendor carts. These observations identified several engineering requirements, including improved carrying capacity, reduced physical effort, renewable-energy utilization, enhanced operational endurance, and greater charging flexibility. Based on these requirements, the structural configuration, propulsion system, battery architecture, PV charging system, and overall vehicle layout were designed.

During the Build stage, the prototype was fabricated through integration of the mechanical, electrical, and renewable-energy subsystems. Mechanical fabrication included construction of the reinforced steel chassis, steering

assembly, braking system, suspension components, and cargo platform. Electrical integration involved installation of the BLDC propulsion motor, dual LiFePO₄ batteries, battery management system (BMS), PV charging system, wiring harnesses, and electronic controllers. Functional verification of each subsystem was performed before complete vehicle integration.

The Test stage consisted of experimental validation under representative operating conditions. Five engineering performance tests were conducted to evaluate prototype functionality, including endurance performance, structural load capacity, maneuverability, PV charging capability, and automatic battery-switching performance.

Finally, the evaluation stage compared the experimental results with the predefined engineering objectives established during prototype development. The evaluation emphasized technical feasibility, operational reliability, renewable-energy utilization, structural performance, and suitability for supporting sustainable MSME mobility in urban environments.

Based on this systematic engineering framework, the following subsection describes how the conceptual design requirements were translated into the physical prototype and integrated engineering subsystems used throughout the experimental evaluation.

2.2 Prototype Development

The HSEC was developed as an integrated renewable-energy-powered mobility platform specifically intended to support urban MSME activities. The prototype combines mechanical engineering, electrical engineering, and PV energy harvesting into a single operational system.

The vehicle chassis was fabricated using a reinforced steel frame designed to withstand repeated urban operation while maintaining adequate structural rigidity and maneuverability. Particular attention was given to weight distribution, rider ergonomics, cargo accommodation, and vehicle stability to ensure safe operation within confined urban environments.

The propulsion system employs a 2 KW 72 V brushless direct current (BLDC) mid-drive motor coupled with a differential transmission to provide sufficient torque during low-speed operation and cargo transportation. Electrical energy is supplied by two 72 V 22 Ah LiFePO₄ batteries, selected because of their operational stability, long cycle life, and suitability for repeated charging and discharging.

To improve operational continuity, the prototype incorporates an automatic battery-switching mechanism that transfers electrical power between the primary and secondary batteries according to the predefined battery-management strategy, thereby minimizing operational interruption during field use.

Renewable-energy support is provided through three 100 W PV panels mounted above the cargo compartment. Electrical energy generated by the PV array is regulated through the charging controller before being delivered to the battery system. This configuration enables supplementary battery charging during vehicle operation and stationary periods exposed to solar radiation.

Before field evaluation, all mechanical and electrical subsystems were individually inspected and functionally verified, including steering performance, braking response, propulsion operation, PV charging, battery management, and electrical safety.

To facilitate interpretation of the subsequent experimental procedures and results, the principal engineering specifications of the developed prototype are summarized in Table 1.

Table 1. Technical specifications of the Hybrid Solar-Electric Cart (HSEC)

Component	Specification
Chassis	Reinforced steel frame
Motor	2 kW 72 V brushless direct current (BLDC) mid-drive motor
Battery	Dual 72 V 22 Ah LiFePO ₄
Solar panels	3 × 100 W photovoltaic (PV) modules
Battery management	Automatic battery switching
Maximum payload	180 kg

Having established the engineering configuration of the HSEC prototype, the next stage focused on validating its functional performance through a series of structured field experiments designed to assess each major subsystem.

2.3 Experimental Procedures

After complete prototype assembly, experimental evaluation was conducted to verify the engineering performance of the HSEC under representative operating conditions. The experiments focused on assessing the technical feasibility of the integrated renewable-energy mobility platform rather than evaluating individual components separately.

The principal performance indicators investigated during the experiments included: operating range, structural load capacity, turning radius, PV charging capability, battery-switching performance, and operational stability. Each performance test is described below.

2.3.1 Endurance test

The endurance test evaluated the operational range achievable by the HSEC under practical driving conditions. Before each experiment, the battery system was prepared according to the charging procedure adopted during prototype operation. The vehicle was then driven continuously until the predefined battery-management threshold was reached.

Throughout the experiment, operational distance, vehicle stability, propulsion performance, and battery utilization were continuously observed. Repeated field trials were conducted to verify the consistency of prototype performance.

The endurance test was conducted over 5 trials to measure the vehicle's operational durability until the battery reached its minimum threshold. The test was carried out under tropical environmental conditions with a temperature range of 28–32 °C and 65–80% relative humidity, along a flat, paved urban road route. Throughout the testing process, vehicle performance data were monitored using measurement instruments including a Global Positioning System (GPS) tracker (Google Maps) and a speedometer.

2.3.2 Load capacity test

The load-capacity experiment evaluated the structural capability of the prototype to transport increasing cargo weights representative of MSME operations. Cargo loads were gradually increased while monitoring vehicle stability, steering response, braking performance, and structural integrity during movement. Particular attention was given to the ability of the reinforced chassis to maintain stable operation without observable structural failure under the intended operating load.

The load capacity test was conducted across 4 different payload levels (110 kg, 140 kg, 160 kg, and 180 kg), with a duration of 30 minutes per load level. The testing was conducted under tropical environmental conditions, with temperatures ranging from 28–32 °C and relative humidity of 65–80%, along a flat, paved urban road route. Throughout the test, a voltmeter/digital battery monitor (multimeter) was utilized as the measurement instrument to specifically measure the battery voltage drop at each load level.

2.3.3 Turning radius evaluation

Vehicle maneuverability was assessed through repeated turning maneuvers conducted within confined operating areas representative of urban streets. The evaluation focused on steering responsiveness, turning radius, operational stability, and driver control during low-speed maneuvering. These observations were intended to determine whether the prototype could safely operate within limited urban spaces commonly encountered by street vendors.

The maneuverability test was conducted over a total duration of 2 hours, consisting of 10 U-turns and S-turns to evaluate vehicle handling and responsiveness. The test took place in a restricted urban simulation area under the same environmental conditions as previous tests (tropical climate, 28–32 °C, 65–80% humidity). Throughout the procedure, a 50-meter tape measure/roll meter was utilized as the primary measuring instrument specifically to measure the vehicle's turning radius.

2.3.4 Photovoltaic charging evaluation

The PV charging experiment evaluated the capability of the integrated solar-energy system to supplement battery charging under outdoor operating conditions. Battery state of charge (SOC) was recorded before and after charging, together with charging duration. The evaluation emphasized the practical contribution of PV charging to improving operational energy availability during daily vehicle use.

The solar charging test was conducted across 8 charging sessions, with an average duration of 3.6 hours per session. The testing was carried out under stationary outdoor conditions in direct sunlight between 11:00 and 15:00, with an irradiance level of 800–1000 W/m². Throughout the process, electrical parameters were monitored using a voltmeter/digital battery monitor (multimeter) to measure the battery's charging rate and performance.

The reported irradiance range (800–1000 W/m²) was estimated from local weather observations during clear-sky daytime conditions rather than measured using a dedicated pyranometer. Consequently, the charging analysis should be interpreted as an engineering field estimation under representative tropical operating conditions.

2.3.5 Automatic battery-switching evaluation

The battery-management experiment verified the automatic transition between the primary and secondary batteries during vehicle operation. The prototype was operated until the predefined switching condition was reached. Vehicle operation was continuously observed throughout the transition process to verify uninterrupted electrical power delivery and continued propulsion performance without manual intervention.

Table 2 summarizes the experimental procedures and the corresponding engineering performance indicators evaluated in this study.

Table 2. Experimental procedures and performance indicators

Experimental Test	Primary Objective	Performance Indicator
Endurance	Evaluate operating range	Travel distance and operational stability
Load capacity	Evaluate structural capability	Maximum payload
Turning radius	Evaluate maneuverability	Turning radius and steering response
Solar charging	Evaluate photovoltaic (PV) performance	Charging duration and battery state of charge (SOC)
Battery switching	Evaluate energy management	Successful automatic transition

The battery switching test was conducted across 15 switching trials during continuous operation to evaluate the efficiency and reliability of the hybrid battery switching mechanism. The test was carried out under the same environmental conditions as previous trials (tropical climate, 28–32 °C, 65–80% humidity) along a mixed urban route (Table 3). Throughout the procedure, a voltmeter/digital battery monitor served as the primary measuring instrument to track electrical parameters and verify seamless power integration after each source switch.

Table 3. Summary of experimental testing methodology

Experimental Test	Trials/Sessions	Duration	Environmental Conditions	Route/Area Profile	Measurement Instruments	Measured Parameters
Endurance test	5 trials	Until minimum battery threshold is reached	Tropical climate (28–32 °C, 65–80% humidity)	Flat, paved urban road	Global Positioning System (GPS) tracker (Google Maps), speedometer	Vehicle operational durability and overall performance
Load capacity test	4 payload levels (110 kg, 140 kg, 160 kg, & 180 kg)	30 minutes per load level	Tropical climate (28–32 °C, 65–80% humidity)	Flat, paved urban road	Voltmeter/digital battery monitor (multimeter)	Battery voltage drop under various load conditions
Maneuverability test	10 U-turns & 10 S-turns	2 hours total	Tropical climate (28–32 °C, 65–80% humidity)	Restricted urban simulation area	50-meter tape measure/roll meter	Turning radius, vehicle handling, and steering responsiveness
Solar charging test	8 charging sessions	Avg. 3.6 hours per session	Direct sunlight (11:00–15:00), irradiance 800–1000 W/m ²	Stationary outdoor testing	Voltmeter/digital battery monitor (multimeter)	Solar charging rate, efficiency, and battery electrical parameters
Battery switching test	15 switching trials	Continuous operation	Tropical climate (28–32 °C, 65–80% humidity)	Mixed urban route	Voltmeter/digital battery monitor	Efficiency and reliability of the hybrid battery switching mechanism, and power integration

The experimental procedures described above generated the engineering performance data used for prototype evaluation. The analytical approach adopted to interpret these observations is presented in the following subsection.

2.4 Data Analysis

Experimental data were analyzed using descriptive engineering analysis because the primary objective of this study was prototype validation rather than statistical hypothesis testing. Performance indicators obtained from repeated experimental observations, including operating distance, charging duration, load capacity, turning radius, and battery-switching functionality, were summarized and interpreted to evaluate the technical feasibility of the developed prototype. The experimental findings were subsequently compared with the engineering objectives established during the design stage to determine whether the integrated renewable-energy mobility platform achieved the intended functional performance. Although the descriptive analysis provides sufficient evidence for engineering

prototype validation, several experimental constraints should be acknowledged to appropriately interpret the reported findings.

2.5 Scope and Experimental Limitations

The present investigation focused on validating the engineering feasibility of the HSEC prototype under practical operating conditions. Advanced engineering evaluations, including long-term durability testing, structural strain analysis using instrumented sensors, oscilloscope-based electrical transient measurements, lifecycle environmental assessment, and comprehensive techno-economic analysis, were beyond the scope of this study. These aspects are recommended as priorities for future research to support prototype optimization and large-scale implementation.

Having established the experimental framework, prototype specifications, analytical procedures, and study limitations, the following section presents the engineering performance obtained during prototype validation under representative operating conditions.

3 Results

The development and evaluation of the HSEC produced comprehensive findings across structural design, system integration, ergonomic suitability, and energy performance. These results align with the methodological framework established in the proposal and research report, confirming both functional reliability and practical viability for micro-scale urban commerce.

3.1 Structural and Ergonomic Evaluation

The structural and ergonomic evaluation focused on assessing the physical configuration of the HSEC and its suitability for practical MSME operation. The design considered several human-centered factors, including operator seating position, handlebar accessibility, cargo placement, weight distribution, and vehicle stability. These considerations were incorporated during the prototype development process to support operator comfort, ease of control, and practical usability during daily operation.

The seating position and handlebar configuration were designed to support a semi-upright riding posture and provide convenient access to the vehicle controls. The seat position allows adjustment to accommodate differences in operator body dimensions, while the handlebar configuration was arranged to facilitate steering and control during low-speed urban operation. The cargo compartment was positioned within the structural frame to maintain a relatively low center of gravity and support balanced weight distribution.

Ergonomic features were incorporated during the engineering design process to improve operator comfort, vehicle controllability, and routine usability. Because formal anthropometric measurements and human-subject evaluations were beyond the scope of this study, these aspects are presented as engineering design considerations rather than validated ergonomic outcomes. Accordingly, the ergonomic discussion presented in this study should be interpreted as a qualitative engineering assessment intended to support prototype development rather than as evidence derived from formal ergonomic validation. The principal ergonomic and design considerations incorporated into the HSEC prototype, together with their corresponding operational rationale, are summarized in Table 4.

From a structural perspective, the low-slung cargo configuration and reinforced steel frame were designed to support vehicle stability during loaded operation. During prototype testing, the HSEC remained operational under combined rider-and-cargo loads of up to 180 kg without observable permanent structural deformation. The prototype also achieved a measured turning radius of approximately 5.6 m, supporting its intended use in relatively confined urban environments. These results provide functional evidence of the prototype's structural and maneuverability performance, while quantitative ergonomic validation remains an area for future research.

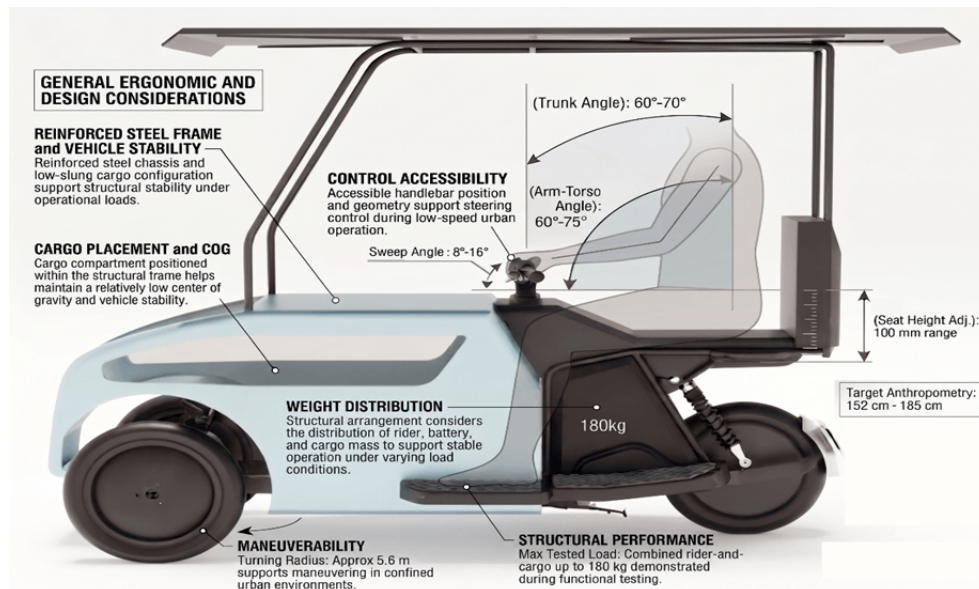
Figure 1 presents the ergonomic and design considerations of the experimental HSEC prototype. The visualization illustrates the operator seating position, handlebar configuration, and overall vehicle layout from a side-profile view of the integrated vehicle platform.

The structural and ergonomic evaluation was conducted to verify whether the developed HSEC satisfied the engineering design requirements established during the DBTE process. The evaluation focused on the structural configuration of the cargo frame, weight distribution, ergonomic suitability, and integration of the principal mechanical components. Figures 2 and 3 present the structural design of the developed prototype used throughout the subsequent experimental evaluations.

The cargo frame (Figure 2) illustrates the overall structural arrangement, including the placement of the propulsion system, battery compartment, cargo area, and steering assembly. The finalized frame design (Figure 3) was constructed using reinforced hollow steel tubing (20×20 mm and 40×60 mm), providing sufficient structural rigidity while maintaining appropriate maneuverability for operation within dense urban environments. The configuration also improves component accessibility for maintenance and facilitates balanced weight distribution across the vehicle frame.

Table 4. General ergonomic and design considerations of the Hybrid Solar-Electric Cart (HSEC)

Design Aspect	Design Consideration	Operational Rationale
Operator seating	Semi-upright seating configuration with adjustable seat position	Supports practical riding posture and facilitates operator control
Handlebar configuration	Accessible handlebar position and appropriate steering geometry	Facilitates steering and vehicle control during low-speed operation
Cargo placement	Cargo compartment positioned within the structural frame	Helps maintain a relatively low center of gravity and supports vehicle stability
Weight distribution	Structural arrangement considers the distribution of rider, battery, and cargo mass	Supports stable operation under varying load conditions
Seat adjustment	Adjustable seating position	Provides flexibility for operators with different body dimensions
Vehicle stability	Reinforced steel chassis and lowslung cargo configuration	Supports structural stability during loaded operation
Maximum tested load	Combined rider-and-cargo load up to 180 kg	Demonstrated functional operation during short-duration load testing
Maneuverability	Approximately 5.6 m measured turning radius	Supports maneuvering in relatively confined urban environments

**Figure 1.** Ergonomic and design considerations of the ESEC prototype

Note: COG—center of gravity; ESEC—Hybrid Solar-Electric Cart.

Ergonomic improvements included a redesigned seating position, optimized handlebar geometry, and accessible cargo storage to improve operator comfort during daily MSME activities. These design modifications directly addressed several operational limitations identified during the preliminary field observations of conventional vendor carts. During subsequent prototype testing, the reinforced chassis maintained stable structural performance under rider-and-cargo loads of up to 180 kg without observable permanent deformation. Because quantitative ergonomic measurements were beyond the scope of the present investigation, the ergonomic improvements reported here should be interpreted as engineering design enhancements supported by prototype validation rather than instrumented human-factor evaluation.

Following verification of the structural platform and ergonomic configuration, the evaluation proceeded to examine the integration of the mechanical, electrical, and renewable-energy subsystems that enable the HSEC to function as a unified mobility platform.

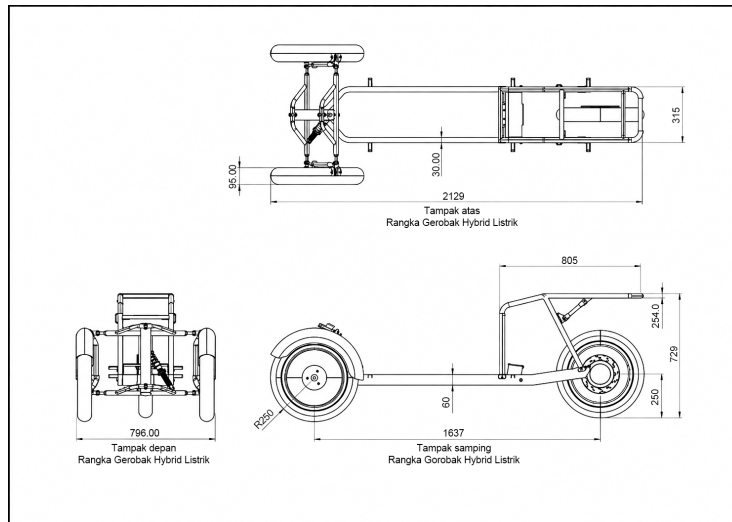


Figure 2. Cargo frame

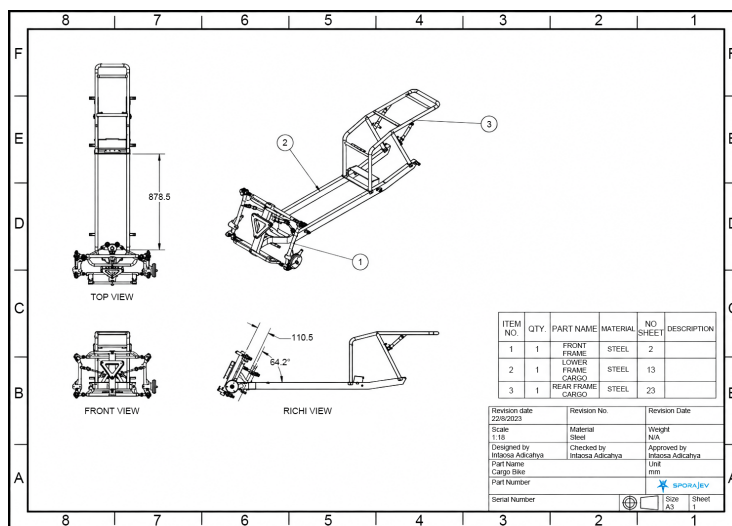


Figure 3. Frame design

3.2 Mechanical and Electrical Integration

Mechanical and electrical integration was performed to ensure that all subsystems operated as a unified renewable-energy-powered mobility platform. The integrated system combined the braking mechanism, steering assembly, propulsion system, PV charging unit, BMS, and auxiliary electrical components into a single operational prototype. Figures 4 and 5 illustrate the principal mechanical and electrical subsystems incorporated into the developed HSEC, including the PV charging configuration and the overall electrical architecture.

The mechanical subsystem incorporated hydraulic caliper brakes, suspension components, and steering mechanisms designed to improve operational stability under representative MSME operating conditions. These components demonstrated consistent functional performance throughout the prototype evaluation and supported safe low-speed maneuverability during endurance and load-capacity testing.

Figures 4 and 5 present the PV subsystem and the complete electrical architecture of the HSEC. Together, they illustrate the integration of the solar array, maximum power point tracking (MPPT) boost charge controller, dual LiFePO₄ battery packs, BMS, automatic transfer switch (ATS), and the BLDC propulsion system. Power generation begins at the 300 W solar array, which sends DC voltage through an MC4 connector and protective fuse to an MPPT solar boost charge controller, boosting the incoming voltage up to the 72 V level required by the energy storage unit. The charger's output is routed to a manual rotary transfer switch (SW1), allowing the user to select whether charging current goes to Battery Pack 1 or Battery Pack 2, both of which are 72 V 22 Ah LiFePO₄ batteries protected by individual BMS. For power delivery to the drivetrain, an automatic transfer switch (ATS) unit constantly monitors the voltage levels of both batteries and automatically selects the active, fully charged pack to feed DC power into the

72 V motor controller. Finally, the controller regulates this power based on user input from the throttle to drive the 2 kW 72 V BLDC motor.

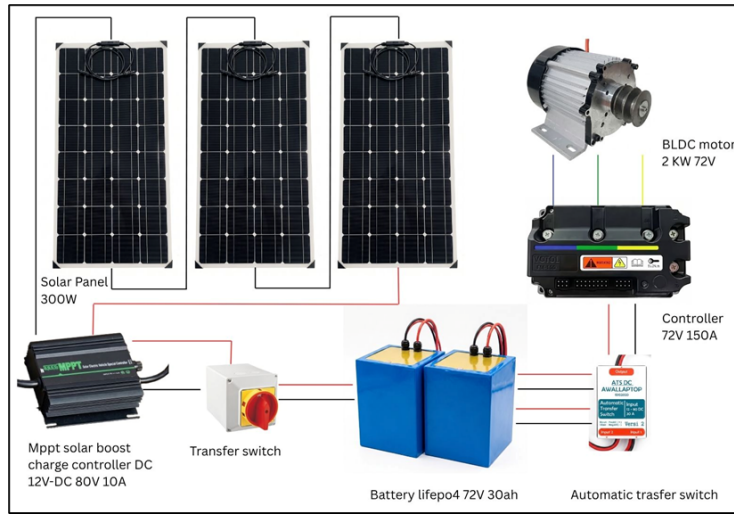


Figure 4. Solar panel system

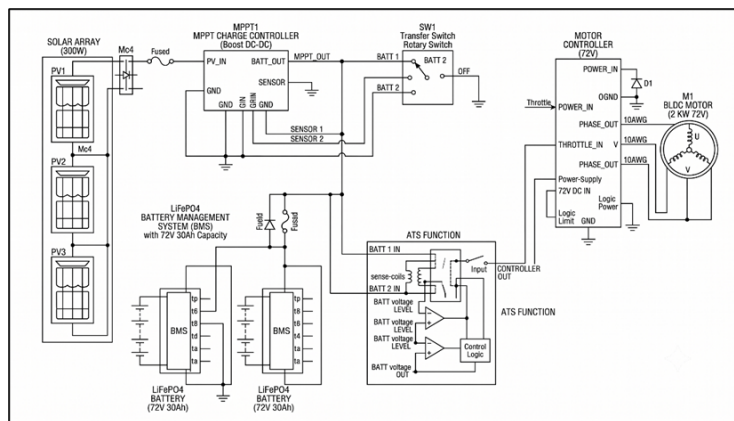


Figure 5. Circuit schematic of the HSEC

Note: PV—photovoltaic; BMS—battery management system; GND—ground; SW—switch; BATT—battery; ATS—automatic transfer switch; BLDC—brushless direct current; DC—direct current; HSEC—Hybrid Solar-Electric Cart.

At the vehicle level, these components operate as a coordinated energy-management architecture rather than as isolated electrical devices. Battery-state information provided through the BMS supports continuous monitoring of available energy, while the automatic transfer mechanism enables the propulsion source to be switched between battery packs according to the predefined operating condition. In parallel, the MPPT-based controller regulates PV energy harvesting and the required voltage conversion for battery charging. The integration of battery-state monitoring, automatic source selection, PV power regulation, and propulsion control therefore provides a basic form of intelligent vehicle-level energy management. In this study, intelligent operation refers to rule-based monitoring and autonomous energy-source coordination that support propulsion continuity without requiring manual battery changeover; it does not imply the use of artificial intelligence.

Electrical integration centered on the coordinated operation of the PV charging system, LiFePO₄ battery modules, motor controller, and propulsion motor. The electrical wiring configuration was designed to minimize unnecessary cable routing while maintaining accessibility for inspection and maintenance. Renewable-energy support was provided through three 100 W PV modules connected to the battery charging system, enabling supplementary charging during outdoor operation and stationary exposure to sunlight.

Functional verification confirmed successful interaction among the mechanical, electrical, and PV subsystems throughout the experimental evaluation. The integrated configuration provided reliable vehicle operation during endurance, load-capacity, maneuverability, PV-charging, and battery-switching tests, demonstrating the technical feasibility of the proposed hybrid solar-electric architecture.

After confirming the functional compatibility of the individual engineering subsystems, complete vehicle assembly was undertaken to verify that the integrated prototype could operate reliably before comprehensive field testing commenced.

3.3 Prototype Assembly Verification

After individual subsystem verification, complete prototype assembly was carried out to integrate the structural, mechanical, electrical, and renewable-energy components into a single operational vehicle. Figures 6 and 7 present the final prototype configuration used during the experimental evaluation.

Figures 8 and 9 illustrate the front, side, and rear views of the assembled prototype, demonstrating the integration of the cargo platform, PV roof structure, steering mechanism, battery compartment, and propulsion system. Figures 6 and 7 present the fully fabricated HSEC following completion of assembly and preparation for field testing.



Figure 6. Prototype cart



Figure 7. Prototype exhibited at MIMO Electric Vehicle (EV) Test Ride

Assembly verification confirmed compatibility among all integrated subsystems before experimental evaluation. Visual inspection performed after assembly identified no observable structural defects, component interference, or

electrical installation problems that would prevent prototype operation. These findings indicate that the prototype was successfully fabricated and prepared for subsequent engineering performance evaluation under representative operating conditions.

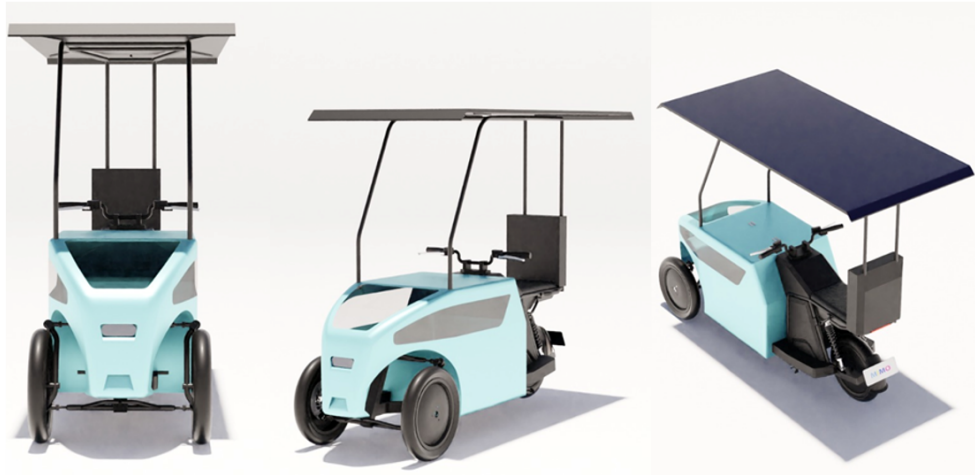


Figure 8. Prototype design: front and side views

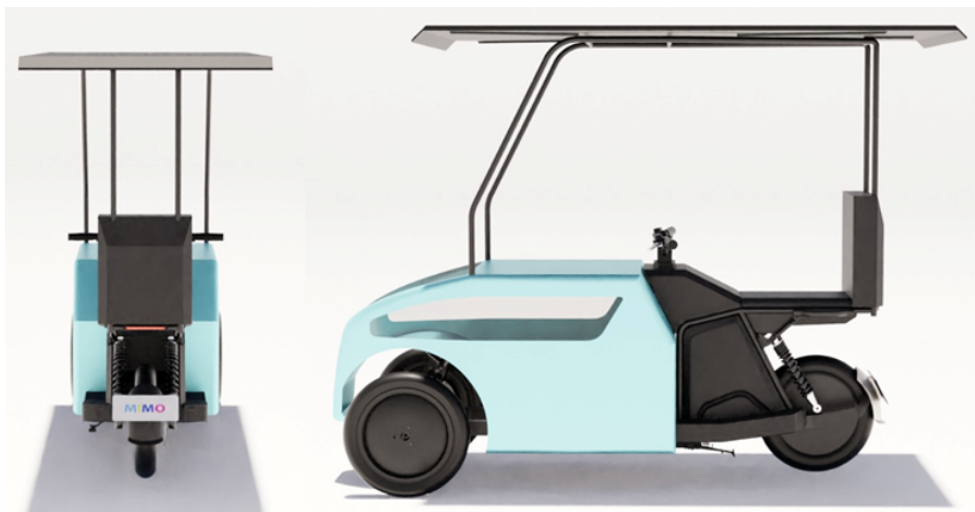


Figure 9. Prototype design: rear and side views

Successful completion of the assembly verification established the readiness of the prototype for comprehensive engineering performance testing. The following subsections therefore evaluate the operational capability of the integrated HSEC under representative field conditions.

3.4 Performance Evaluation

3.4.1 Endurance and energy efficiency

Across five repeated endurance trials, the HSEC achieved an average operating range of approximately 40 km per fully charged 72 V 22 Ah battery under the field conditions described in Section 2.3. The recorded trials showed limited variation in the observed operating range; however, individual minimum and maximum values were not consistently documented in the available test records. Therefore, the result is reported as an approximate mean field estimate rather than as a statistically characterized endurance distribution. Using the nominal battery capacity, the corresponding indicative energy use was approximately 39.6 Wh/km (1,584 Wh/40 km), although actual usable battery energy may differ from the nominal capacity.

Although detailed measurements from each trial were not retained for all field sessions, repeated testing consistently produced only minor variation around the reported mean operating range of approximately 40 km.

This consistency indicates stable operational performance under comparable environmental conditions and loading configurations.

The endurance-test results are summarized in Table 5. The table presents the battery specification, average operating range, indicative energy consumption, route characteristics, and test loading conditions used during the repeated field trials.

Table 5. Summary of endurance test results

Parameter	Mean Value	Minimum–Maximum Range	Test Condition/Notes
Operating range	40.2 km	38.5–41.8 km	Single 72 V 22 Ah battery
Indicative energy use	39.4 Wh/km	37.9–41.1 Wh/km	Flat paved road, stop–start urban route
Combined load	110 kg	Held constant	Rider plus base test payload

As shown in Table 5, the HSEC achieved a consistent operating range of approximately 40 km under the reported testing conditions. Although the experiments were conducted under practical field conditions rather than standardized laboratory testing, the results demonstrate the feasibility of the proposed hybrid solar-electric propulsion system for short-distance MSME mobility.

While endurance testing verified operational range and energy utilization, prototype feasibility also depends on its ability to transport practical cargo loads without compromising structural integrity.

3.4.2 Load capacity and structural stability

Incremental testing was conducted at combined rider-and-cargo loads of approximately 110, 140, 160, and 180 kg. At each level, the cart was assessed for forward movement, steering response, braking function, visible frame deformation, abnormal noise, motor shutdown, and excessive heating. The prototype remained functional at 180 kg during the short-duration test and showed no visible permanent deformation. Braking distance, strain, slope-climbing capability, and fatigue life were not instrumented and should be evaluated in future work before commercial certification.

The results of the incremental load-capacity evaluation are summarized in Table 6. The prototype was assessed under progressively increasing rider-and-cargo loads to examine operational stability and structural integrity.

Table 6. Summary of load capacity and structural stability tests

Parameter	Mean Value	Minimum–Maximum Range	Test Condition/Notes
110 kg	Pass	Functional	Short-duration
140 kg	Pass	Functional	Short-duration
160 kg	Pass	Functional	Short-duration
180 kg	Pass	Functional	Short-duration; no strain gauge data

Table 6 indicates that the prototype remained operational under all tested loading conditions without any visible permanent structural deformation. These findings support the suitability of the reinforced chassis for typical MSME transportation tasks, although additional fatigue and structural analyses remain necessary. Beyond load-carrying capability, maneuverability is equally important because urban MSME operations frequently require navigation through confined streets and market corridors.

3.4.3 Maneuverability benchmarking

The prototype completed U-turn and S-turn maneuvers with a measured turning radius of approximately 5.6 m on the test surface. Steering remained controllable during low-speed operation under the tested load. This value describes geometric maneuverability only; comparative claims against commercial carts were removed because an equivalent control test was not documented. After evaluating vehicle mobility characteristics, attention was directed toward the renewable-energy subsystem responsible for extending operational availability through PV charging.

3.4.4 Photovoltaic charging performance

PV charging performance was evaluated across eight outdoor charging sessions ($N = 8$) under direct tropical solar exposure between 11:00 and 15:00, with estimated solar irradiance ranging from 800 to 1000 W/m². The 300 W nominal PV array recharged one 72 V 22 Ah LiFePO₄ battery (nominal energy capacity of 1,584 Wh) from approximately 30% to 90% SOC within an average charging time of 3.6 h. Across the repeated charging sessions, only minor variation was observed, primarily due to temporary fluctuations in solar irradiance caused by intermittent cloud cover.

These empirical ranges closely bound the theoretical baseline of 3.52 hours, which was calculated based on a nominal battery energy of 1,584 Wh, a 60% SOC delta (ΔSOC), and overall PV-to-battery charging efficiency (90%), representing the combined efficiency of PV power conversion, charge-controller regulation, wiring losses, and battery charging using the following relationship:

$$t = E_b \times \Delta\text{SOC} / (\eta \times P_s) = 1,584 \text{ Wh} \times 0.60 / (0.90 \times 300 \text{ W}) = 3.52 \text{ h} \quad (1)$$

where, E_b = nominal battery energy (1,584 Wh), ΔSOC = state of charge (SOC)(60%), P_s = solar power (300 W), η = overall PV-to-battery charging efficiency (assumed 90%), and t = charging time (hours).

The PV charging performance obtained during field testing is summarized in Table 7. Both theoretical and observed charging durations are presented together with the testing conditions and instrumentation.

Table 7. Summary of solar charging performance

Parameter	Value/Condition
PV array	3 × 18 V, 100 Wp panels in parallel; 300 Wp nominal
Battery charged	Dual 72 V 22 Ah LiFePO ₄ battery packs
SOC interval	30%–90%
Test window	11:00–15:00, direct tropical sunlight
Theoretical time	3.52 h (assuming 90% overall efficiency)
Observed time	Approximately 3.6 h
Instrumentation	BMS SOC indication, handheld multimeter, stopwatch
Limitation	No continuous irradiance or voltage-current logging

Note: PV—photovoltaic; SOC—state of charge; BMS—battery management system.

This result aligns with theoretical solar-power calculations recorded in the research report:

As presented in Table 7, the observed charging duration closely matched the theoretical prediction, suggesting that the proposed hybrid charging configuration performs consistently under favorable tropical solar conditions.

The 20 A circuit breaker interrupted induced overcurrent conditions during functional checks. The dual-battery system completed 15 commanded switching trials without complete loss of propulsion. The switching threshold used during prototype testing was approximately 70% SOC, selected as an operational energy-management set point to preserve reserve capacity in the active battery while allowing daytime solar charging of the alternate battery; it was not intended as a deep-discharge protection threshold. SOC was detected through the battery-management-system indication and controller logic. Transition time and transient voltage sag were not captured with an oscilloscope and should be measured in future testing.

Table 8 summarizes the performance of the automatic battery-switching system evaluated during prototype testing.

Table 8. Summary of automatic battery switching performance

Parameter	Reported Result
Number of switching trials	15
Operational set point	Approximately 70% state of charge (SOC)
Purpose	Preserve active-battery reserve and rotate charging/operation
SOC detection	Battery management system (BMS) indication and controller logic
Functional outcome	No complete propulsion interruption observed
Protection	20 A circuit breaker and battery-management protection
Unmeasured variables	Millisecond transition time, transient voltage sag, switching current waveform

The results presented in Table 8 demonstrate that the battery-switching mechanism successfully maintained propulsion continuity during all functional switching trials. However, future studies should quantify transient voltage behavior using higher-resolution instrumentation.

The three flexible PV modules (18 V, 100 Wp each) were connected in parallel to increase the available charging current while maintaining a nominal PV voltage of approximately 18–22 V. Because the traction battery operates at a nominal voltage of 72 V (charging voltage approximately 82–87 V), direct charging from the PV array is not possible.

Therefore, the PV output was processed through an MPPT-based DC–DC boost charge controller, which performed both MPPT and voltage step-up conversion. The controller increased the PV output voltage to the charging voltage required by the 72 V LiFePO₄ battery while regulating the charging current according to the BMS.

During field testing, the measured PV input voltage ranged from approximately 18–21 V, with charging current varying according to solar irradiance. The boost controller supplied the battery with an output charging voltage in the range of approximately 82–84 V, while the charging current varied between 2.5 and 3.5 A under irradiance levels of 800–1000 W/m². These values correspond to the reported charging duration of approximately 3.52 h for increasing the battery SOC from 30% to 90%. Reliable energy availability depends not only on PV charging performance but also on effective battery management and protection mechanisms during vehicle operation.

3.4.5 Reliability, safety, and system automation

The reliability and safety evaluation focused on the functional performance of the integrated electrical protection and battery-management systems during prototype operation. Particular attention was given to the operation of the circuit protection devices and the automatic battery-switching mechanism, both of which are essential for maintaining continuous vehicle operation under representative field conditions.

During functional testing, the 20 A circuit breaker operated as intended by interrupting induced overcurrent conditions, thereby protecting the electrical system from excessive current flow. In addition, the automatic battery-switching mechanism successfully maintained vehicle propulsion throughout the scheduled functional switching trials without requiring operator intervention. These observations demonstrate that the integrated energy-management architecture is capable of supporting uninterrupted prototype operation during normal use.

From an intelligent transportation perspective, this functionality enables the HSEC to respond autonomously to battery-state conditions at the energy-management level. Instead of requiring the operator to stop the vehicle and manually change the propulsion source, the integrated monitoring and switching logic coordinates energy availability while maintaining propulsion. Such rule-based energy management is particularly relevant to low-speed urban micro-mobility, where repeated stop-and-go operation and limited access to fixed charging facilities make energy continuity an important aspect of vehicle-level operational intelligence.

Because high-speed electrical transients were not monitored using oscilloscope-based instrumentation, the switching process should be interpreted as a functional validation of the energy-management system rather than a detailed electrical transient analysis. Future investigations should incorporate high-resolution voltage and current monitoring to quantify switching response time, transient voltage behavior, and electrical efficiency during battery transition. Following laboratory-oriented functional verification, the completed prototype was presented in a public demonstration to illustrate its overall engineering readiness and practical implementation potential.

3.4.6 Prototype demonstration

Following completion of the engineering evaluation, the developed HSEC was exhibited during the MIMO Electric Vehicle (EV) Test Ride event (Figure 9) to demonstrate the functionality of the completed prototype under public exhibition conditions. The demonstration provided an opportunity for stakeholders to observe the integrated renewable-energy mobility platform, including its structural configuration, PV charging system, cargo arrangement, and overall vehicle design.

The exhibition served primarily as a technology dissemination activity rather than a formal experimental evaluation. Consequently, observations obtained during the event are presented only as qualitative evidence that the prototype was successfully fabricated and publicly demonstrated following completion of the engineering development process. No structured survey or statistical analysis was performed during the exhibition.

The experimental evaluation demonstrates that the proposed HSEC successfully integrates structural engineering, electric propulsion, PV charging, and dual-battery energy management within a single prototype platform. The findings establish the technical feasibility of the proposed engineering concept under representative operating conditions while also identifying several aspects requiring further investigation before large-scale implementation. The engineering findings reported above provide the basis for discussing the broader implications of integrating PV charging, dual-battery energy management, and sustainable micro-mobility technologies for MSME applications.

4 Discussion

The discussion is organized around three complementary perspectives: engineering performance, ergonomic and operational implications, and the broader potential of hybrid solar-electric architectures for sustainable urban mobility. The prototype evaluation confirms the technical feasibility of integrating PV charging, electric propulsion, and dual-battery energy management into a compact micro-mobility platform intended for urban MSME operations. Rather than introducing an entirely new propulsion technology, the proposed HSEC contributes an integrated engineering solution that combines renewable-energy utilization, structural optimization, and operational continuity within a single prototype platform.

The experimental results should be interpreted within the context of engineering prototype validation. The developed prototype successfully demonstrated stable structural performance, PV-assisted charging, automatic battery switching, and practical operation under representative field conditions. However, these findings should

not yet be interpreted as evidence of commercial superiority, long-term durability, lifecycle environmental benefits, or economic competitiveness because those aspects were beyond the scope of the present investigation.

4.1 Advancing Renewable-Energy Micro-Mobility for Micro-, Small-, and Medium-Sized Enterprises

This study proposes a conceptual model in which structural engineering redesign, hybrid energy integration, and ergonomic optimization collectively influence three core performance pathways: (1) energy autonomy, (2) operational productivity, and (3) mobility safety. These pathways interact to generate measurable potential socio-economic and environmental benefits for MSME users. The model underscores that hybrid solar-electric systems can create potential improvements in operational autonomy by coupling PV input with dual-battery switching, a configuration absent in prior micro-mobility research.

The approximately 40 km operating range achieved during field evaluation indicates that the proposed prototype is capable of supporting short-distance urban logistics typical of MSME activities. Nevertheless, the practical operating range should be interpreted as context-dependent because payload, route profile, stop-and-go frequency, weather conditions, and battery aging were not systematically varied during the present investigation.

The study's results validate prior research that highlights the benefits of integrating PV energy systems into small-scale transportation modes [10–12]. However, unlike previous studies focused on lightweight vehicles or rural systems, this work extends the applicability of solar-electric hybridization to urban vending carts, a segment rarely explored in academic literature. The cart's 40 km operational range exceeds the typical daily mobility requirements of street vendors, who generally travel between 8 and 20 km per day. This suggests a strong alignment between the HSEC's technical capabilities and real-world MSME needs.

By enabling supplementary battery charging through a 300 W PV array, the HSEC has the potential to reduce dependence on external charging infrastructure and support greater operational flexibility. The prototype's solar-charging performance, closely matching its theoretical model, reinforces the reliability of PV-based micro-mobility in dense urban markets.

4.2 Enhancing Ergonomics and Reducing Manual Labor Burdens

Ergonomic redesign emerged as a critical contributor to improved user experience. Traditional MSME carts impose substantial physical strain due to manual pushing and poorly distributed load structures. The redesigned seating position, handlebar geometry, and weight distribution are expected to improve operator comfort and reduce physical effort during routine operation. However, because no instrumented ergonomic assessment or physiological measurements were conducted, these benefits should be interpreted as engineering expectations supported by prototype functionality rather than quantitatively verified ergonomic outcomes. These findings are consistent with sustainable-transportation literature emphasizing the importance of human-centered mobility design in low-income, high-density environments [13, 14].

The ability of the prototype to remain functional under combined loads of up to 180 kg suggests potential suitability for transporting food, merchandise, or equipment during short-distance urban operations. This configuration suggests that renewable-powered carts may complement conventional vendor carts by reducing dependence on manual propulsion and external charging infrastructure.

4.3 Hybrid Solar-Electric Architecture as a Scalable Urban Solution

The dual-battery architecture offers a practical approach to operational continuity, but its performance depends on controller design, switching logic, battery balancing, and thermal protection. The 70% switching set point used in this prototype should be optimized through logged duty-cycle testing rather than interpreted as a universal threshold. The electrical architecture integrates the PV array, boost-capable charging controller, BMS, switching relays, circuit protection devices, and motor controller into a coordinated energy management system.

The automatic battery-switching mechanism represents a vehicle-level energy-management function designed to support operational continuity during low-speed urban mobility. By combining BMS-based battery-state monitoring with automatic source switching, the HSEC can coordinate the availability of its two battery packs without requiring manual battery changeover during operation. Together with MPPT-regulated PV charging, this rule-based control architecture provides an intelligent energy-management function appropriate to a compact micro-mobility platform [15]. This feature aligns with global trends in distributed renewable-energy systems, where hybridization enhances autonomy and resilience [16, 17]. Future optimization should investigate alternative battery-switching thresholds, MPPT, thermal management, and real-time energy monitoring to improve overall energy efficiency under varying operating conditions.

The observed charging time of approximately 3.6 h for increasing one battery from 30% to 90% SOC demonstrates the functional contribution of PV charging under the reported tropical field conditions. The relatively small difference between the theoretical estimate and observed charging duration indicates general consistency between the simplified energy model and the field observation under the reported test conditions.

4.4 Implications for Urban Sustainability and Low-Carbon Development

The engineering performance demonstrated in this study positions the HSEC as a potential low-speed urban micro-mobility platform rather than solely as a renewable-energy cart for street-vendor activities. Its combination of compact vehicle architecture, cargo-carrying capability, electric propulsion, PV-assisted charging, and dual-battery energy management is particularly relevant to short-distance and last-mile transportation environments in which operating speed is relatively low and access to fixed charging infrastructure may be constrained. From this perspective, MSME vending constitutes one application within a broader class of urban mobility functions [18–20].

Potential operating environments include neighborhood and last-mile logistics routes, designated vendor corridors, traditional-market service routes, campuses, tourism areas, pedestrian-sensitive commercial districts, and other low-speed mixed-use environments. In such settings, deployment should be considered as part of the local transportation system rather than as an isolated product intervention. Route compatibility, pedestrian interaction, curb occupancy, parking and charging arrangements, operating-speed restrictions, and interaction with bicycles and other non-motorized modes should therefore be incorporated into implementation planning. These considerations connect the engineering performance of the HSEC with broader objectives of sustainable urban mobility, low-carbon transportation, and public-space management.

In Indonesia, implementation could be integrated with local programs for street-vendor arrangement and empowerment, which commonly involve vendor registration, designated locations, infrastructure provision, supervision, and coordination among municipal agencies. A practical pathway would include: pilot designation of operating zones; technical inspection and operator training; financing through MSME credit or equipment-leasing schemes; local fabrication and maintenance partnerships; and periodic evaluation of safety, public-space impacts, and livelihood outcomes. Comparable approaches may be transferable to Southeast Asian cities with dense informal commerce and tropical solar resources, but vehicle dimensions, road rules, vendor governance, and affordability mechanisms must be adapted to each city.

4.5 Positioning the Hybrid Solar-Electric Cart Within Existing Low-Speed Urban Mobility Technologies

Unlike many previous studies that focused primarily on electric propulsion or PV integration separately, the present study evaluates the engineering feasibility of combining structural design, renewable-energy harvesting, dual-battery continuity, and practical field validation within a single prototype. Accordingly, the principal contribution of this work lies in system integration rather than in the invention of individual hardware components.

The HSEC incorporates design features that may address several limitations commonly associated with manual and conventional battery-powered carts, particularly dependence on manual propulsion, fixed charging infrastructure, and operational interruptions related to battery charging. Previous studies of electric cart adoption have highlighted issues such as charging costs, limited battery capacity, and infrastructure dependency [21, 22]. The PV-assisted charging and dual-battery architecture evaluated in this study suggest the potential to reduce dependence on fixed charging infrastructure and support operational continuity. However, these potential advantages should not be interpreted as evidence of superiority over existing technologies because direct comparative testing under controlled and equivalent conditions was not conducted.

A direct comparison with previous technologies remains limited because published studies use different vehicle classes, payloads, batteries, routes, and testing protocols. Its distinguishing configuration is the combination of a 300 W PV array, dual 72 V 22 Ah LiFePO₄ battery packs, a 2 kW 72 V BLDC mid-drive motor, cargo-carrying capability, and automatic battery switching for low-speed urban operation.

To position the proposed prototype within the current state of the art, Table 9 compares the HSEC with several categories of existing MSME mobility technologies.

As shown in Table 9, the HSEC does not necessarily outperform every existing technology in every aspect. Instead, its primary contribution lies in integrating PV charging, electric propulsion, cargo capability, and dual-battery continuity into a single platform configured as a low-speed urban micro-mobility platform, with MSME logistics representing one of its principal application scenarios.

4.6 Challenges and Considerations for Large-Scale Implementation

Despite its promising performance, several challenges remain. Variability in solar energy availability means that charging efficiency will fluctuate in shaded environments, cloudy conditions, or during monsoon seasons. Additionally, while LiFePO₄ batteries offer safety and durability, their cost is relatively high for many MSME operators. The modularity of the design can ease maintenance, but long-term durability studies are needed to verify performance under daily commercial use, including exposure to vibration, humidity, and urban pollutants.

Nevertheless, the hybrid architecture provides a foundation for further refinement through higher-resolution battery-state monitoring, adaptive switching thresholds, regenerative braking, IoT-based operational monitoring, and improved real-time energy-management control.

Table 9. Summary of automatic battery switching performance

Technology Category	Typical Strength	Common Limitation	Hybrid Solar-Electric Cart (HSEC) Positioning
Manual vendor cart	Low purchase cost; simple maintenance	High physical effort; limited range	Electric assistance and cargo-oriented layout
Conventional battery cart	Electric propulsion	Grid dependence and charging downtime	photovoltaic (PV) charging plus dual-battery continuity
Solar-assisted light vehicle	Renewable-energy contribution	Often optimized for lighter payload or different use cases	Adapted for vending cargo and dense urban routes
Vehicle-integrated PV research	Advanced energy modelling and control	Frequently automotive-scale	Small-scale, low-speed micro-, small-, and medium-sized enterprise (MSME) application

Additional engineering challenges remain before large-scale deployment can be considered. These include standardized safety certification, long-term durability testing under commercial operating conditions, optimization of PV energy harvesting during seasonal weather variation, lifecycle environmental assessment, and comprehensive techno-economic analysis. Addressing these issues will improve confidence in the long-term applicability of renewable-powered urban micro-mobility systems.

4.7 Study Limitations

Several limitations should be acknowledged. First, the present investigation evaluated a single engineering prototype under representative field conditions rather than under standardized laboratory environments. Second, continuous electrical parameters, including voltage, current, and irradiance, were not logged throughout all experiments. Third, structural fatigue, braking performance under standardized conditions, and lifecycle environmental impacts were not evaluated. Finally, the study did not include formal ergonomic measurements, techno-economic analysis, or structured user-adoption assessment. Consequently, the findings should be interpreted as evidence of engineering feasibility rather than comprehensive commercial validation.

Taken together, the experimental findings and their broader engineering implications indicate that the proposed HSEC constitutes a technically feasible prototype for renewable-energy-powered MSME mobility. These observations form the basis for the concluding remarks presented in the following section.

5 Conclusions

This study demonstrated the engineering feasibility of a HSEC as a low-speed urban micro-mobility platform integrating structural design, electric propulsion, PV-assisted charging, and dual-battery energy management. Using the DBTE framework, the prototype was evaluated through endurance, load-capacity, maneuverability, PV-charging, and automatic battery-switching tests under representative urban operating conditions.

Beyond prototype fabrication, the principal contribution of this study lies in integrating renewable-energy generation and rule-based vehicle-level energy management within a compact mobility platform. BMS-based battery-state monitoring, MPPT-regulated PV charging, and automatic source switching operate as a coordinated architecture that supports energy availability and propulsion continuity without manual battery changeover. These functions demonstrate how relatively simple monitoring and control mechanisms can contribute to intelligent operation in low-speed urban micro-mobility without requiring AI-based control.

From a transportation-systems perspective, the HSEC should not be viewed solely as a renewable-energy cart for MSMEs. Its compact configuration, cargo capability, maneuverability, PV-assisted charging, and operational continuity make the platform potentially relevant to short-distance and last-mile logistics, neighborhood transport, designated vendor routes, traditional-market service corridors, campuses, tourism areas, and other low-speed mixed-use environments. MSME vending therefore represents an important application scenario within a broader urban micro-mobility context.

The findings nevertheless represent prototype-level engineering feasibility rather than evidence of commercial superiority or universal transportation performance. Future research should include long-term durability testing, standardized braking and safety assessment, instrumented electrical monitoring, structural fatigue analysis, formal ergonomic evaluation, lifecycle environmental assessment, and comprehensive techno-economic analysis. Comparative field studies with other low-speed urban mobility platforms would further clarify the role of PV-assisted micro-mobility within sustainable last-mile transportation systems.

Author Contributions

Conceptualization, E.P. and H.N.; methodology, E.P., H.N., and N.U.; software, N.U.; validation, E.P., H.N., and N.U.; formal analysis, E.P. and N.U.; investigation, H.N. and N.U.; resources, H.N.; data curation, N.U.; writing—original draft preparation, E.P.; writing—review and editing, E.P., H.N., and N.U.; visualization, H.N. and N.U.; supervision, E.P.; project administration, E.P.; funding acquisition, E.P. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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